

Logging infrastructure recommendation

Audience: Product / operations. Purpose: agree a safe logging approach for carland_auth, carland_service, webhook, and future services — under current host limits and zero tooling budget.

Date context: production host already runs the application stack.

1. Current situation

- Services: carland_auth, carland_service, webhook (+ future services).
- Host: DigitalOcean Basic — 1 vCPU / 2 GB RAM / 50 GB disk.
- Budget for logging tooling: none (no paid managed plan, no second server).
- Primary need: investigate errors and reconstruct “what happened”.
- Not required yet: long-term compliance archive, full-text analytics at scale.

Constraint: the same 2 GB host also runs application containers and supporting infrastructure. Any heavy logging stack competes directly with product traffic.

2. What was evaluated

- Elasticsearch + Logstash/Filebeat + Kibana (ELK) — strong search UX, high RAM cost.
- Filebeat → Elasticsearch → Kibana without Logstash — still ES-heavy.
- Grafana Loki + Promtail — lighter than ES, still non-trivial on 2 GB.
- Free-tier managed log clouds — possible later; quotas and retention limits apply.
- Local Docker/container logs + rotation — lowest resource cost.

3. Recommendation (now)

Do not deploy Elasticsearch / full ELK / large Loki on the current 2 GB host. Risk: memory pressure, OOM kills, downtime for auth/service/webhook.

Adopt a phased approach:

Phase 0 (now, \$0, same host)

- Structured application logs (JSON) to container stdout
- Short retention via log rotation (e.g. 3–7 days) + compression
- Shared fields: service, env, level, timestamp, request/trace id
- Never log secrets, selfie/livePhoto, raw FIN, HMAC keys

Phase 1 (still \$0, optional)

- Ship ERROR/WARN only to a free-tier cloud log product (e.g. Grafana Cloud Loki free) if quota allows
- Keep debug/info local and short-lived

Phase 2 (when budget or capacity exists)

- Separate small logging host, or RAM upgrade (recommended ≥ 4 –8 GB), or paid managed logging — then ELK/Loki becomes viable

4. Why not ELK on this host?

- Elasticsearch alone often needs ~1–2 GB RAM — most of the machine.
- Apps + DB/redis/proxy + ES on 2 GB is operationally unsafe.
- 50 GB disk fills quickly if retention is long and logs are verbose.
- ELK remains a valid future architecture — wrong fit for Phase 0.

5. Design principles (all phases)

- One shape for all services: consistent JSON fields and levels.
- Correlation: request id / trace id across auth, service, webhook.
- Privacy by default: mask phone/FIN; never store biometric payloads.
- Prefer stdout over ad-hoc log files (simpler ops under Docker).
- Retention explicit: hot window short; archive only when capacity exists.
- Volume control: sample or drop noisy INFO in production if needed.

6. “Where can we send logs for free?”

There is no unlimited free off-box archive. Options under zero budget: *CarCat — Logging infrastructure recommendation*

- Same host, rotated local logs — safest for stability (recommended default).
- Free-tier managed logging — “aside” storage with monthly GB / retention caps; suitable for errors only, not full debug history.
- True long-term central logging requires either a second host, a RAM upgrade, or a paid plan later.

7. Decision summary for Product

Ask now:

Accept Phase 0 (local structured logs + short retention)?

Optional:

Trial free-tier ERROR shipping (Phase 1) without installing ES locally?

Later investment trigger:

Need multi-week search across all services → upgrade host or buy managed logging.

8. Outcome if we follow this

- Production stability preserved on the current 2 GB droplet.
- Engineers can still investigate recent failures via rotated logs.
- Clear upgrade path when budget or incident volume requires central search.
- Avoids building a logging stack that competes with customer-facing services.

Internal recommendation — CarCat platform. Regenerate: python docs/generate_logging_infrastructure_recommendation_pdf.py